8 (a) (i) State what is meant by nuclear binding energy.

.....

.....

..... [2]

(ii) On Fig. 8.1, sketch a line to show the variation with nucleon number A of the binding energy per nucleon E of a nucleus.

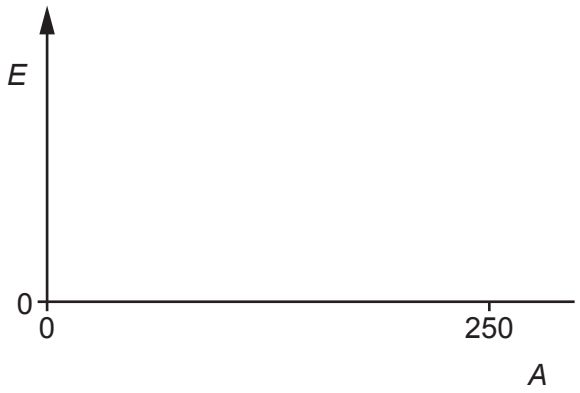


Fig. 8.1

[2]

(b) In one type of nuclear process, deuterium ($^{2}_{1}H$) undergoes the reaction

$$^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{3}_{2}He + {}^{1}_{0}n.$$

(i) State the name of this type of nuclear process.

..... [1]

(ii) Explain, with reference to your line in **(a)(ii)**, why this reaction results in the release of energy.

.....

.....

..... [2]

(c) Table 8.1 shows the masses of the particles involved in the reaction in **(b)**.

Table 8.1

particle	mass / u
$^{1}_{0}n$	1.008 665
$^{2}_{1}H$	2.014 102
$^{3}_{2}He$	3.016 029

Calculate the energy released when 1.00 mol of deuterium undergoes the reaction.

energy = ... J [5]

[Total: 12]